

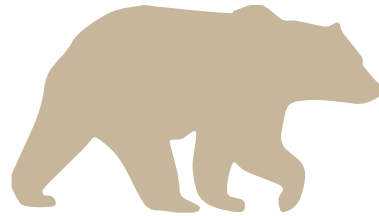


# BearLodge

Find your next home



# TABLE OF CONTENTS



|    |                                |
|----|--------------------------------|
| 2  | INTRODUCTION                   |
| 3  | BRAINSTORMING                  |
| 4  | TASK ANALYSIS                  |
| 5  | INTERMEDIATE SKETCHES          |
| 6  | STAKEHOLDER INTERVIEW FINDINGS |
| 7  | SCENARIOS                      |
| 9  | INITIAL DRAFT                  |
| 10 | WIREFRAME                      |
| 11 | USER TESTING                   |
| 13 | FINAL DESIGN                   |
| 15 | API                            |
| 17 | TECHNICAL CHALLENGES           |
| 19 | CONCLUSION                     |
| 19 | APPENDIX A                     |

## Introducing BearLodge

For our final project, we decided to create BearLodge, a modern housing app for Berkeley students to find their new homes. While creating BearLodge, we gathered feedback from current students on their experience with searching for housing in Berkeley. A common thread from all of our interviews was that most students need to use a plethora of different methods to find a housing option that suits them. Using this feedback, we designed a sleek and modern user experience that makes searching for housing safe and simple for young Bears.



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# Brainstorming

Our team came up with **25 unique ideas** before zeroing in on the idea that later became BearLodge. With the general theme of deconstruction and reconstruction in mind, our team brainstormed a diverse set of ideas, ranging in topic from housing to music to dating apps. Here are some of our favorite brainstormed ideas:



**Running and Spotify Beats-** utilizes an ML algorithm to match the beat and tempo of songs that to the pace of running or workouts. Deconstructs the process of creating a workout playlist and reconstructs it to be more beneficial and match the energy of workouts.



**Library seating application-** This is a live display of available seats in a library that can be accessed by users. Deconstructs the physical atmosphere of looking for available seating (which can be difficult during Finals), and reconstructs it to a virtual setting and makes the process easier.



**Gender-neutral bathrooms:** deconstruct the process of searching for gender-neutral bathrooms on campus and reconstruct into a map that displays where all the bathrooms are and how to access them

And finally, the idea that we ended up developing into **BearLodge:**

**Housing application-** This is an aggregation of housing pages (facebook groups, slack, networking for roommates, apartment applications), **a one stop app** for Berkeley students to find roommates/ share leases. It deconstructs the disorganized way students try to find housing and **combats Berkeley's current housing crisis**. Then reconstructs it to be more organized and safer from scams on facebook pages by making it **Berkeley affiliated** and making sure everyone on the platform is **accredited** (ex. Has a Berkeley email)

## Theme

BearLodge **deconstructs the oft-complex process** of finding housing in Berkeley; namely the steps in the process such as online listing browsing, Facebook group joining, or networking with peers that can be overwhelming for busy and/or inexperienced students. BearLodge then **reconstructs this process into one simple and safe user experience**, offering account verification that ensures all buyers are confirmed Berkeley students.

# Task Analysis

After some consideration, we realized that there were three main scenarios and user groups that we wanted to cater to with our housing app: Berkeley students who are looking for housing, Berkeley students who have a space to offer other students, and third-party landlords and sellers looking for students to live in their spaces. We performed a task analysis on the current way that these three user groups are currently meeting their housing needs:



## Looking for housing as a Berkeley student:

- 1.Look at various postings
  - a.Look on Facebook Housing groups
  - b.Asking friends/word of mouth
  - c.Posting on club Slacks
  - d.Searching Zillow (other housing websites)
  - e.Asking on Wechat
- 2.Message subletter
  - a.Exchange contact information
  - b.Negotiate listings and offers
- 3.Verify safety/credibility of lister
  - a.Check renter's Housing License
  - b.Check that they are an accredited student of Berkeley



## Looking for subletters as a Berkeley student:

- 1.Post in Facebook Housing groups, Slack #random channels, ask friends
  - a.Upload Pictures/ video tours /information about in-person tours
  - b.Write descriptions
  - c.Amenities, Utilities, Parking, address, pets
  - d.Directly message possible renters
  - e.Post with contact information
- 2.Message possible renter/subletter
  - a.Answer questions regarding pricing, utilities, location
  - b.Ask questions -- how COVID-safe are you, security deposit, terms of lease
- 3.Reach agreement
  - a.Getting them in touch with landlord
  - b.Contact information
  - c.Sign the lease

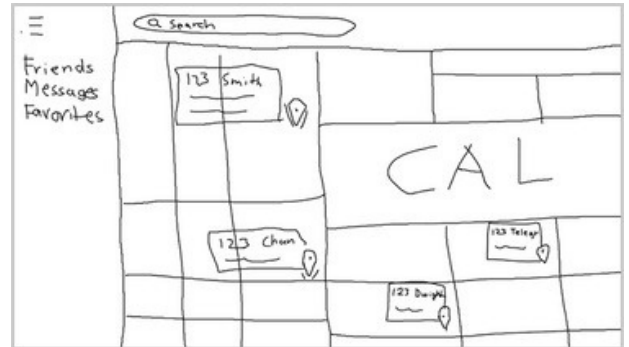
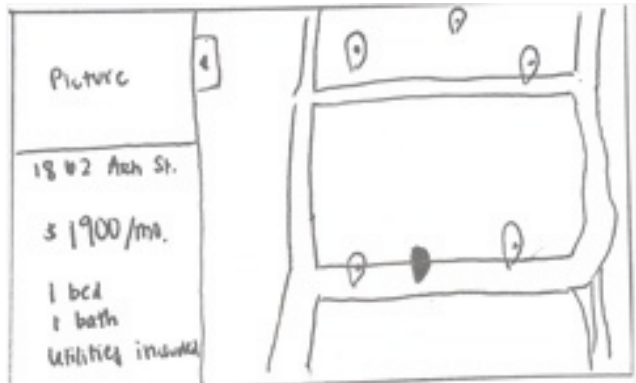


## Looking for renters/subletters as an external party:

- 1.Advertise listings online
  - a.Facebook Housing groups
  - b.Put it on personal website
  - c.Put it up on Zillow, etc.
  - d.Post with contact information
- 2.Message possible renter
  - a.Answer questions regarding pricing, utilities, location
  - b.Ask questions -- how COVID-safe are you, security deposit, terms of lease
- 3.Agreement reached
  - a.Getting them in touch with landlord
  - b.Contact information
  - c.Sign the lease

# Intermediate Sketches

Sketching was helpful because it allowed us to further refine our project idea. This stage in the design process took our app from a series of tasks determined through the task analysis into a set of screens and interactions. By having everyone on the team contribute to these sketches, we were able to consider a wide range of design possibilities before proceeding with the wireframe.



LANDING PAGE:



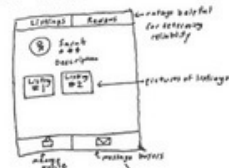
Register:  
app company name  
learning activity  
or school name

Forgot:  
Ask for school  
Email

BUYER VIEW:



SELLER'S NEW PROFILE:





## Stakeholder Interview Findings

We noticed a few recurring patterns across the 6 interviews that we conducted. The first was **having to visit many different platforms to meet their housing needs**. 4 out of 6 of our interviewed participants reporting having to use 2 or more avenues such as Facebook, Zillow, WeChat, and UC Berkeley's roommate portal in order to meet their housing needs, whether that was finding somewhere to live, finding roommates, or finding renters for an apartment they were renting out. A second concern shared among many of our participants was **evading scams or verifying the legitimacy of their potential roommate or landlord**. Participants reported having to "look through everything with eagle eyes", only going through friend's recommendation, or scheduling tours with sellers in order to verify sellers/roommates and evade scams.

This feedback told us that verifying the legitimacy of all users and listings would be an important feature of our application, and would let us become a one-stop-shop for students to find housing, so that surfing various different platforms would no longer be necessary.

## Final Idea

BearLodge is a mobile application to find and advertise housing! It can be difficult to navigate Berkeley housing, especially as someone who is renting their first apartment or is new to the area. This process often involves juggling Facebook groups, Slack messages, and informal conversations with those who are looking to sublet. The status quo is fragmented and deconstructed, and BearLodge seeks to fix that. As a platform for Berkeley students to connect with one another, we consolidate everything into one place, reconstructing the apartment-hunting process. It is a convenient single location for buyers and sellers alike. BearLodge: Find Your Next Home!

And **BearLodge was born!**



# Slack Feedback

Although we weren't able to receive too much feedback on our Slack post, this stage was still helpful because we went through the process of writing an app description and questions for feedback. In coming up with a description for our app, we were able to succinctly describe the intended user experience and interactions that we wanted to design, and the feedback questions narrowed the focus to the parts of the app that we were most interested in/uncertain about.

## Feedback questions:

- Are we missing any important features?
- On a scale of 1-5 (1 is very unimportant and 5 is very important) how important are each of the following features to delivering on the mission of the app?
  - Map that displays listings
  - In-app messaging between buyers and sellers
  - Reviews of sellers
  - Authenticating users as Cal students (e.g. using Google OAuth)
- What parts of the design are unclear/excessive/etc.?
- Any other general comments/questions/suggestions!



## Scenarios

The personas and scenarios were helpful to sharpen our understanding of our users, especially since there are three distinct scenarios/user groups that were catering toward: students looking for housing, students looking for other student subletters, and landlords looking for student tenants. It was also helpful to pinpoint the current pain points with other avenues our users may be using today to meet their needs, like facebook and zillow.

Annie

Sadie

Jeff



### Berk student tenant:

Persona: Annie is a sophomore at Berkeley who has no experience looking for off campus housing. She wants to find an apartment to live in for the upcoming semester, and is looking for other students to live with. She's only ever lived in the dorms before, and wants to make sure that she secures housing through a trustworthy source.



### Scenario:

She decides to join various Facebook groups and look on websites such as Zillow, and Trulia. With all the different ongoing conversations, it's hard to organize and remember which houses she likes, and from which websites they are from. In addition to this, she almost gets caught in a scam from an untrustworthy company. She searches for an official school affiliated app. Upon registering, the app verifies that she is a Berkeley student and she feels much safer using this app to contact potential roommates and subletters. With all the listings and messages in one platform, she is quickly able to compare different leases and find a trustworthy listing.

### Berk student looking for subletter:

Sadie just completed her third year at Cal. Although she was planning on just graduating one semester early, in the fall, she realized that she could just complete her remaining two courses over the summer and graduate even earlier than expected. However, now she needs to find someone to take over her lease for the fall semester. She wants to make sure to sublet her apartment to somebody trustworthy, and would prefer to find a fellow student to take over the lease.



### Scenario:

She comes up with a nice description of her apartment on Southside, takes a few photos, and posts it on UC Berkeley Housing, Rooms, Apartments, and Sublets Facebook group. Soon after, she receives a few Facebook messages from prospective tenants. Some are not very interested and are just asking things that are already detailed in the post, but others are genuinely interested and want to do a walk-through. However, it's difficult filtering through the spam, and staying on top of her messages is a lot of work for her busy schedule. In addition, she has to verify that each prospective tenant is a student and not a scammer by looking for their social media profiles or asking mutual friends. One of her friends tells her about a school-affiliated app where every person looking for housing is a verified Berkeley student. She makes a posting on the app and messages with prospective tenants, and is able to find a junior, Angelina, who would like to sublet from Sadie.

## External landlord looking for renter:

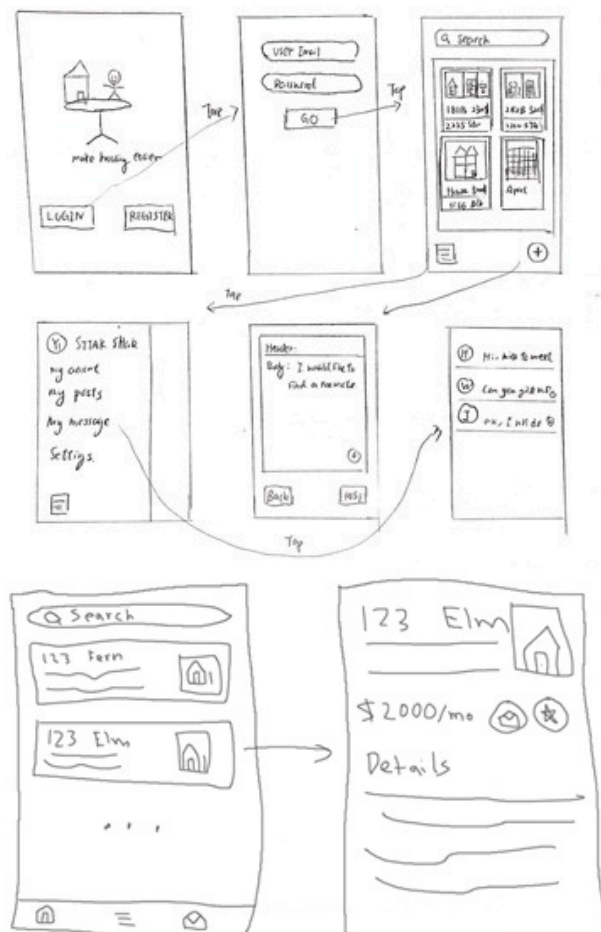
Jeff is a landlord in Berkeley. He owns about a dozen properties in an apartment complex on Northside, close to Soda Hall. Ever since the pandemic, he's had a lot of trouble finding renters. Most of his previous tenants have been UC Berkeley students, so several of his properties have been vacant since March 2020. He's excited that Berkeley announced in-person classes in the fall, and he wants to advertise his vacant properties to potential tenants.



### Scenario:

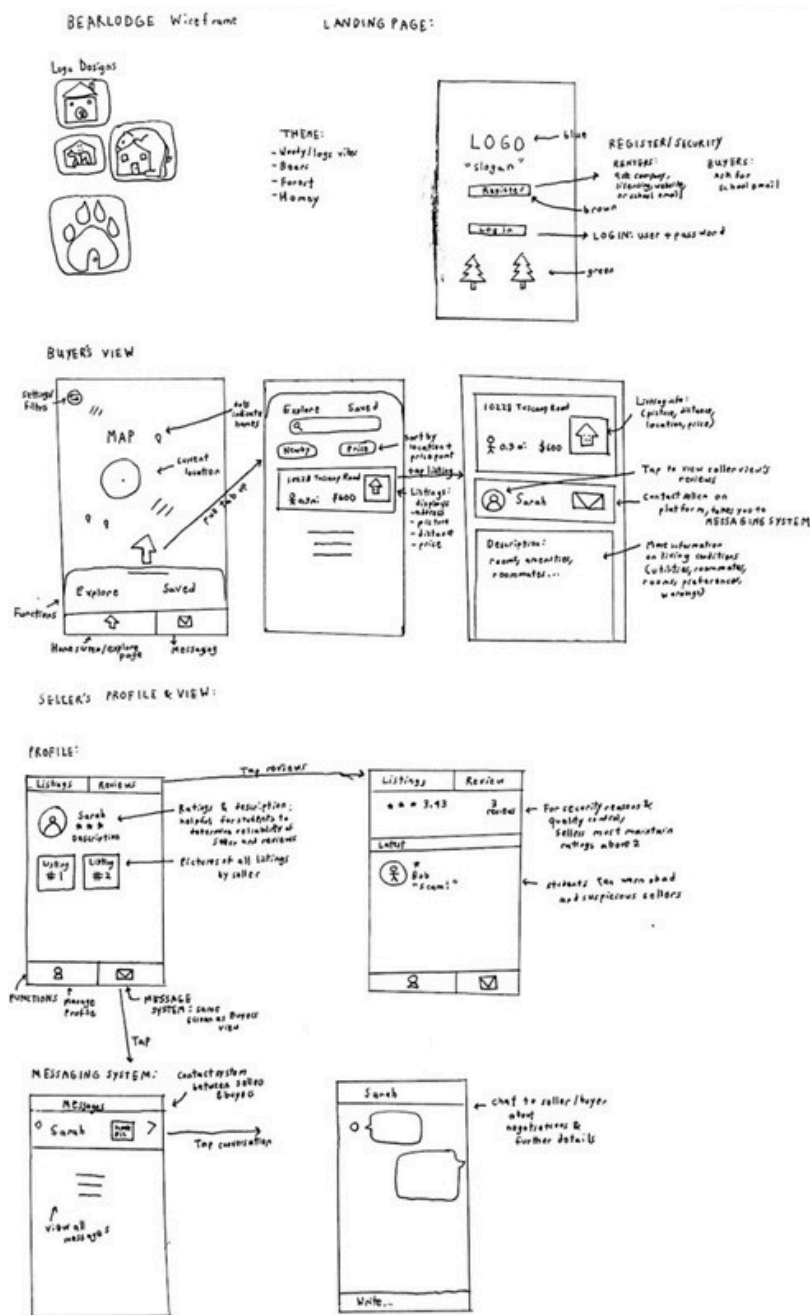
Jeff pulls out the descriptions he has written for each of his units, updates them with some minor changes (lowering the rent and security deposit amounts), and he posts them on a Berkeley-affiliated housing app he found. In a few days, he receives a message from a junior, David, who's interested in learning more about the property. After some back and forth, they arrange a walk-through of the property. Ultimately, David and his roommate, Alexis, sign a rental agreement.

## Initial Draft



# Wireframe

Creating the wireframe was the most essential part of implementing the tasks by integrating the users and tasks of the app to a design that would be beneficial to both buyers and sellers. Integrating feedback from the interviews and my teammates, I was able to create an architectural representation of a user interface in order to test functionality and determine priority of content. It was useful to flesh out ideas, and determine control flow with arrows, and determine the basic graphics and theme.





# User Testing

The user-free evaluation helped us understand how a user would interact with each feature within our app and it exposed some interactions that we hadn't mapped. For example, when a user clicks 'contact renter', we noticed that the Figma prototype had this leading them to the list of all their direct messages (their inbox). Instead of this, we realized that we should have this button go directly to the direct messaging screen with the seller. Overall, though, we felt that much of the cognitive walkthrough felt intuitive and was already considered in other parts of the design process.

**User Free Evaluation- Cognitive Walkthrough**

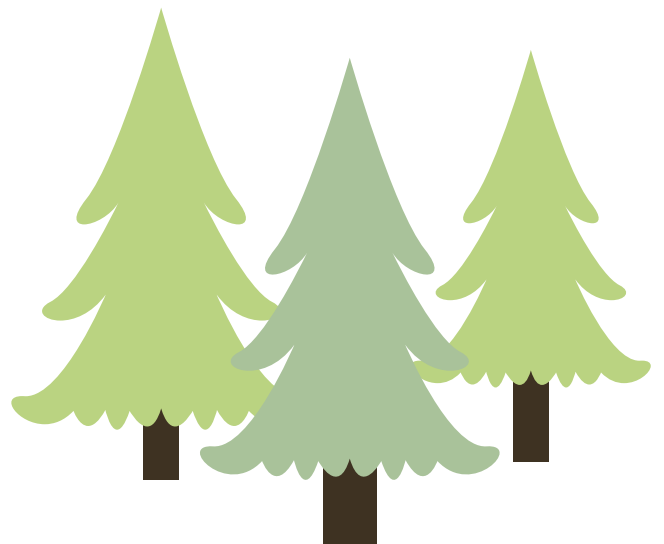
| Action                                                           | What outcome is the user trying to achieve, and is it the intended goal of the system                                                                                                                    | Will the user notice that the correct action is available to them?                                                                                         | Will the user associate the correct action with the outcome they expect to achieve?                                  | If the correct action is performed; will the user see (on the interface) that progress is being made towards their intended outcome?                                 |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Action 1:</b> User clicks for more information about a house. | See more information about a certain listing (either opening a context pop-up, or another window with more information). Yes, it's the intended goal.                                                    | Potentially not. It isn't explicitly stated anywhere that clicking on the card opens a view with more information on that property.                        | Yes, users who click on the card are expecting more info, and that's what they'll get.                               | Yes, a new view with information will open.                                                                                                                          |
| <b>Action 2:</b> User clicks to contact renter                   | They're likely expecting either a newly opened chat between them and the renter, or some method to send a first message.. That's the intended goal of the interaction.                                   | Yes! The contact renter button is highlighted by its location on the screen, and its accent color.                                                         | Yes, the interaction is well-labeled and highlighted so the user interaction will be well defined before it is done. | Yes! A view change will occur to the messaging system that allows the user to contact the renter.                                                                    |
| <b>Action 3:</b> User clicks to choose a renter to chat          | Similar to the property exploration screen, ongoing chats are displayed in a list where clicking on a given chat card opens the specific chat.                                                           | Again, it isn't stated anywhere explicitly that clicking a chat card opens the corresponding chat window, but many users will expect that behavior anyway. | When users click the card, they expect the chat to open, and that's what happens.                                    | Yes, the view will change to the contextual chat window.                                                                                                             |
| <b>Action 4:</b> User clicks to see messaging system             | The icon for the messaging system is a mail icon in the bottom right. When the user clicks the mail icon, they want to see messages, or an inbox of some sort, which is the intended goal of the system. | Yes, and the icon is even highlighted when new messages have been received.                                                                                | Yes, the icon for messages is extremely self explanatory.                                                            | Yes. Instead of whatever view the user came from (could be any, as the icon is on a fixed footer taskbar), they will be presented with a list of available messages. |

# User Evaluation

## Questions:

- 1.What are your initial thoughts on the landing screen?
- 2.Let's imagine a scenario where you want to use this app to look for housing:  
walk me through how you would do this.
- 3.What do you think would happen if you click edit listing?
- 4.How would you use the Explore Page?
- 5.Which parts of the interface are most confusing?
- 6.What strikes you as different about the Student vs. Renter's View?
- 7.Would any additional features or functionality be helpful?

**Insights:** Overall, the users we tested with enjoyed the interactions of our website. The general design was well received, with a few minor suggestions offered regarding coloring or additional instructional labelling on how to get started. The users generally intuitively navigated through creating an account and getting started with the website. The most common confusing area for the users was the seller view: not having a more intuitive way to recognize that the listings were clickable, that the view was not as fleshed out as the buyer view, etc. Also, many of the users suggested that another key functionality to be added would be additional filtering options for the explore page. Being able to filter better by location, price, listing characteristics (number of rooms, etc.) would be extremely helpful to a selective college student.



# Final Design

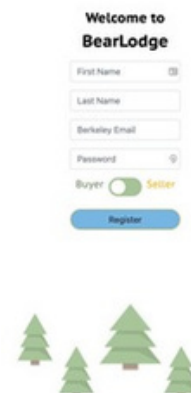
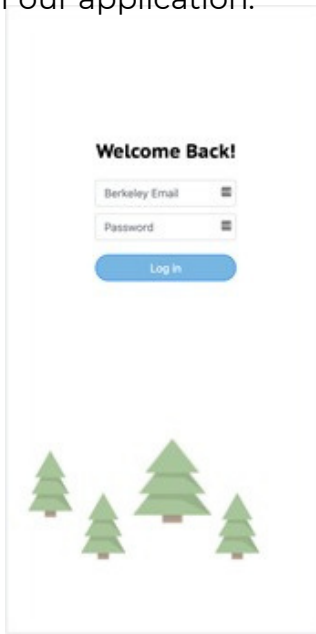
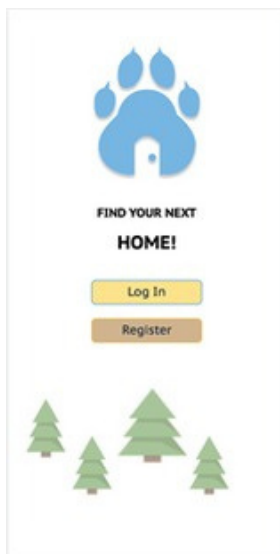
The high-fidelity prototype allowed for real time end user interaction which allowed us to test the look and feel of the prototype. Creating this prototype helped us map out the details, the integration of buttons and was extremely helpful for creating a design and helping guide the code architecture for the final product.



## Features



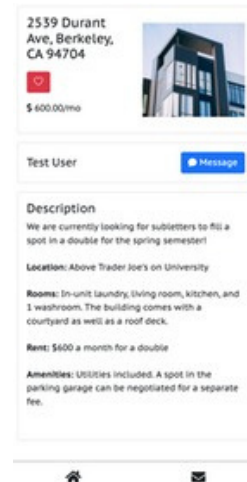
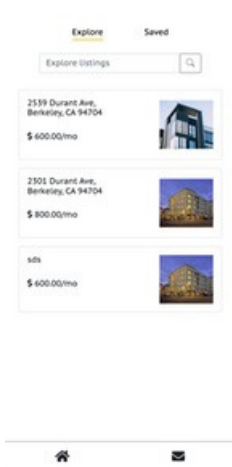
1. **Log In and Verification** - In order to store user information (which listings they've saved, which ones they've posted), we created user login and registration screens. Moreover, our app is meant to be specifically for Berkeley students, so during the registration process, we verify users as such. This verification process made this task a critical component of our application.



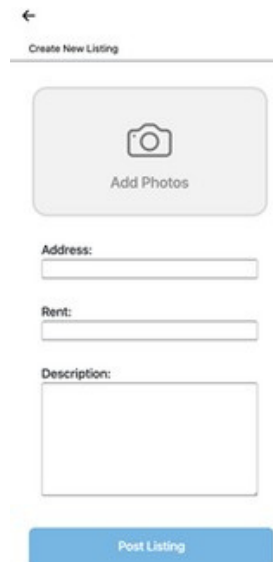
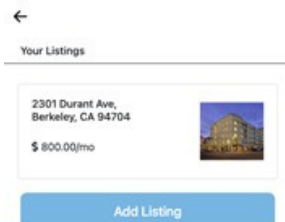




**2. Explore Listings and Save Listings** - Users are able to see a list of all available listings. They are also able to search for listings nearby a query location. Moreover, they can tap into a listing to see the full description of it, and in this detailed view, they can save the listing, so it appears on a separate page.



**3. Create and Edit Listings** - Users who register as owners can view a list of all their listings as well as create new listings and edit existing ones.



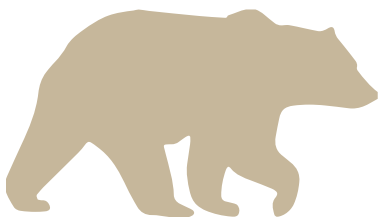
## FIGMA:

<https://www.figma.com/file/M4VFHwPfigvcKGfEcvkVhb/BearLodge?node-id=0%3A1>

## API

This part of the design process helped us understand what will be needed from a technical perspective to implement our Django prototype. We started out big-picture, and made sure that we understood all the technologies (APIs, database tables, etc.) that we would need to implement the application that was in our Figma high-fidelity prototype. Then, we evaluated how much work each component of the prototype would take to implement and moved some features (like CalCentral authentication and reviews) out of scope for the MVP, for the sake of getting all the work completed by the deadline.

/ Home screen - log in or register,  
/log\_in Log in screen  
/register Registration screen



### CalCentral Auth - home screen

To implement authentication of the user as a Cal student, we investigated using CalCentral. One example we found that does this is Berkeleytime. Berkeleytime has users log in to save their profile information when they are building their class schedules. It uses Google OAuth to have users sign in with their Berkeley GSuite credentials and then verifies that the user is part of the berkeley.edu domain. Here is a link to Berkeleytime's GitHub where they do this: <https://github.com/asuc-octo/berkeleytime/blob/ecfbaddbc84551e2617e34990520281f87408267/backend/user/views.py>.

After discussing among team members, we decided that using Google OAuth might be too time consuming for the minimal value that it adds to the application. So, we decided against using this service. Instead, we decided to go with Django's out-of-the-box authentication solution: <https://docs.djangoproject.com/en/3.2/topics/auth/>.



/explore User searches for location and it shows nearby listings.

/explore/{listing\_id} Listing detail page

/saved Displays the user's saved listings

/listings Seller's view of all their listings

/listings/create Seller's view to add a new listing

/listings/{listing\_id}/edit Seller view to edit listing



/messages Screen that displays all messages

/messages/{user\_id} Message with a specific user



## Map

One view in our application includes a map that displays all available listings that match the user's search criteria. In order to implement this, we investigated various JavaScript libraries that help with mapping. After doing some research, we decided between MapBox and Google Maps. Both have extensive documentation and support online. We decided to go with MapBox because it is extremely customizable and has a visually appealing UI.

**Messaging** In our task analysis and application prototype, we have included a feature where potential tenants and renters are able to message one another to discuss the details of listings and set up walk-throughs. One option for implementing this feature is to base the UI off Bootstrap like this example on JSFiddle. Then, when a user writes a message to another user, we store the message in our database. We also poll the database to look for any new messages; when we find them, we insert them into a container in the view as HTML elements using jQuery. We would use the overflow property in Bootstrap to get the messages to vertically scroll. If this is difficult to maintain in real-time, we may also want to open a websocket, so the user is able to listen for and send messages and persist messages in our database layer.

This feature was also assessed to be quite a bit of work. Yongxi decided to take this on. If after working on this feature for a bit, he decides it is more work than expected, we will move it out of scope.

## **Database Design**

### Profile

- user (1-1 with Users)
- saved\_listings (many-to-many with Listings)
- is\_seller - Boolean

### Messages

- from\_user (many-to-1 with Users)
- to\_user (many-to-1 with Users)
- sent\_time - Timestamp
- body - String

### Listings

- seller (many-to-1 with Users)
- monthly\_price - Double
- latitude - Long decimal
- longitude - Long decimal
- description - String

image\_url - String

## **Database Solution**

In order to store user information, like profile information, listings, and direct messages between users, we need some database solution. Above, we've designed the database schema as a relational database, so these are the options that we are looking into. After reviewing the codebase from project 4, we noticed that SQLite is the database engine that was already set up in the project. In order to keep things simple, we decided to keep this solution in our project.

# **Technical Challenges**

## **Django**

Since our team was relatively new to Django, ramping up on using the web framework presented a challenge. During the development process, we needed to learn how to use Django to interact with the SQLite database engine. This included creating, editing, and querying rental listings.

## **Front End**

To implement our front-end, we used Django's templating system along with Bootstrap as the CSS framework. Although the team worked off of the same Figma high-fidelity prototype to develop the front-end, there were still some inconsistencies between the screens. Because the development process was so rapid, we did not have the opportunity to build reusable UI components, so particular styling of similar components (e.g. cards, headers, forms) was slightly different. Moreover, since our team was new to Django, we were not very familiar with its HTML templating options and this contributed to our learning curve.



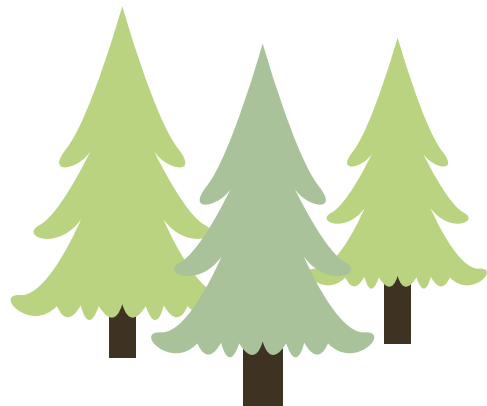
**Authentication and Verification** One pain point for our target users that we identified through interviews was spam and a lack of trust in the credibility of advertisements. We aimed to tackle this pain point in our application, and we did this by restricting our app's users to only Berkeley students, in order to ensure it remained a safe space. Creating this safe space required that we implement an authentication and verification system. We used Django's default authentication solution to register and log in users as well as verify that their email addresses end in 'berkeley.edu.' Learning how to use this authentication mechanism represented another technical challenge for our team, but we were ultimately successful in using it.

### **Geocoding**

Through user interviews, we identified that location is an important factor that Berkeley students use when looking for housing. Therefore, we knew it was important that users be able to query an address and find listings near it. In terms of technical implementation, this required storing the latitude and longitude coordinates of each listing in the database. One challenge here was translating user-input addresses to coordinates. This process is known as geocoding, and thankfully, we were able to use an API called Positionstack to accomplish this. After getting the query location's coordinates, we needed to sort listings by proximity to these coordinates. For long-term scalability, we identified that using Django's GeoDjango plug-in may be the best way to handle and query geographic information. But to quickly roll out our MVP, we simply calculated the Euclidean of each listing from the query location and then sorted listings in order of least to greatest distance.

### **Messaging**

In Milestone 2 of the design process, we identified the technical requirements that would be necessary to implement our high-fidelity prototype. In this process, we also assessed the work required to implement each feature and decide whether or not to keep the feature within scope of our Django implementation. After assessing in-app messaging, we determined that it would be a significant amount of work and we would keep it as a stretch goal for the team. As anticipated, in-app messaging was a significant challenge to implement, especially keeping the messaging real-time and designing the unique chat screen. Therefore, we decided to keep in-app messaging out of the final application.



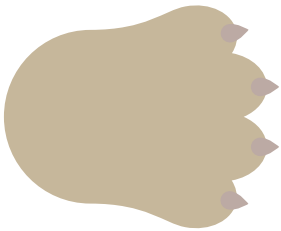
# Conclusion

The complexity of the process to find housing can be a serious challenge for Berkeley students, especially those who may have never lived alone before. With BearLodge, we hope to bring the collective Berkeley housing community into the modern age, and offer a smooth user experience for our fellow students. In the future, we hope to expand to other college campuses so that our work can benefit other student communities.

**In the future:** We hope that the implementation of our app can:

- **Ensure trusted users** with extra measures taken.
- **Easily and Safely** browse homes.
- **Streamline** the process in one-place.

# Appendix A



**Github Repository:** <https://github.com/saahilchadha1/BearLodge>

## Slides:

[https://docs.google.com/presentation/d/1URxeSkwNQvN34ModNvFBpdEkWt8dK7f8G2IV\\_2ORi9I/edit?usp=sharing](https://docs.google.com/presentation/d/1URxeSkwNQvN34ModNvFBpdEkWt8dK7f8G2IV_2ORi9I/edit?usp=sharing)

**Final Video:** <https://youtu.be/cwNXIGMFyn0>



# BEARLODGE

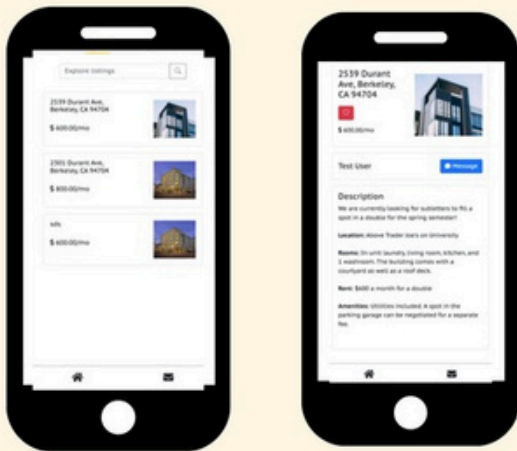
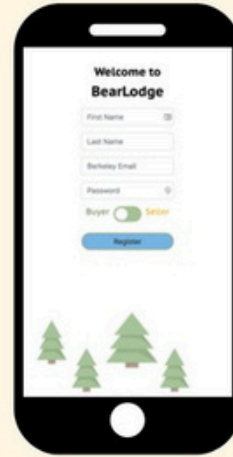
Find your next home.



## The Housing Crisis

We're a group of Berkeley students seeking to create a modern housing app helping Berkeley Bears find their dens

WHO WE ARE



TASK 01

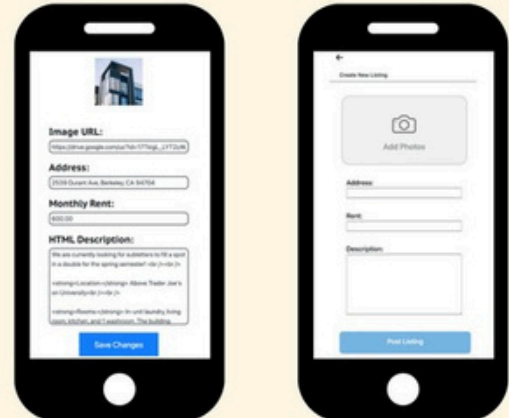
## Log In & Verification

We verify all user's affiliation with the school to improve the security and trustworthiness of our app.

TASK 02

## Explore & Save Listings

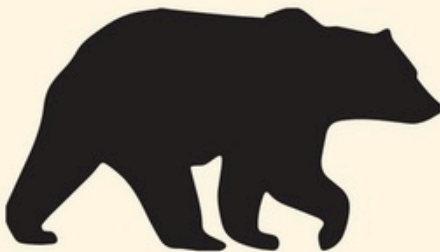
Users can query locations for listings and click for full descriptions. They can save a listing for easy future access.



TASK 03

## Manage Listings

For users who register as owners. They can maintain their own listings as well as add and edit old listings.



## Student Community

We hope to deconstruct students' disorganized process of apartment hunting, and create a one stop app that is reliable and safe.

OUR IMPACT

